

To evaluate the accuracy of the phylogenetic relationships inferred from the mitochondrial cytochrome c oxidase subunit I (COX1) gene, the resulting phylogenetic tree was compared with the phylogenomic study of Li et al. (2016), which reconstructed the evolutionary history of extant Felidae using genome-wide nuclear DNA, complete mitochondrial genomes, and sex chromosome markers. Unlike the present study, which relied solely on a single mitochondrial marker (COX1), Li et al. (2016) employed a comprehensive genomic dataset to resolve evolutionary relationships and investigate the effects of historical hybridization among felid lineages.

The COX1 phylogenetic tree recovered two well-supported monophyletic clades corresponding to the genera *Felis* and *Panthera*. The *Felis* clade comprised *Felis chaus*, *Felis silvestris*, and *Felis catus*, whereas the *Panthera* clade consisted of *Panthera tigris*, *Panthera uncia*, and *Panthera pardus*. This topology is consistent with the phylogeny presented by Li et al. (2016), which recognized both genera as distinct evolutionary lineages within the family Felidae. The placement of *Vulpes vulpes* outside the Felidae clade further supports the use of the red fox as an appropriate outgroup for rooting the phylogenetic tree.

Within the *Felis* clade, the present study recovered *Felis silvestris* and *Felis catus* as sister taxa, supported by a local support value of 0.994, while *Felis chaus* diverged earlier within the clade. This relationship agrees with Li et al. (2016), who identified domestic cats and wildcats as members of the Domestic Cat lineage and demonstrated their close evolutionary affinity using genome-scale molecular data. The concordance between the COX1 phylogeny and the genome-wide analysis indicates that the mitochondrial COX1 gene effectively resolves evolutionary relationships among closely related members of the genus *Felis*.

Similarly, the *Panthera* clade was recovered as a distinct monophyletic lineage. In the present analysis, *Panthera uncia* and *Panthera pardus* formed a sister group with a support value of 0.859, whereas *Panthera tigris* diverged earlier within the clade. This overall grouping agrees with Li et al. (2016), which consistently recognized *Panthera* as a natural evolutionary lineage. However, Li et al. (2016) reported that the internal relationships among *Panthera* species may vary depending on whether mitochondrial or nuclear genomic datasets are analyzed because ancient hybridization and incomplete lineage sorting have influenced the evolutionary history of the genus. Consequently, slight differences in the branching order among *Panthera* species between the two studies are expected.

The observed similarities indicate that the mitochondrial COX1 gene retains sufficient phylogenetic signal to recover the major evolutionary divisions within Felidae, particularly at the genus level. Nevertheless, several methodological differences explain the minor discrepancies between the present study and the published phylogeny. First, the current analysis utilized only a single mitochondrial locus inherited maternally, whereas Li et al. (2016) integrated information from genome-wide nuclear markers, complete mitochondrial genomes, and sex chromosomes, providing substantially greater phylogenetic resolution. Second, the published study included 38 extant felid species, while the present analysis examined only six felid species and one canid outgroup, limiting taxonomic representation. Finally, Li et al. (2016) demonstrated that historical

hybridization among ancestral felid lineages contributed to discordance between mitochondrial and nuclear phylogenies, a phenomenon that cannot be fully resolved using a single mitochondrial marker.

Overall, the COX1-based phylogenetic tree generated in the present study is highly consistent with the comprehensive phylogenomic framework proposed by Li et al. (2016). Both studies support the separation of *Felis* and *Panthera* into distinct monophyletic lineages and the close relationship between *Felis catus* and *Felis silvestris*. The minor differences observed within the *Panthera* clade most likely reflect the greater resolving power of genome-wide datasets compared with a single mitochondrial gene. Thus, despite its limited genetic scope, the COX1 marker proved effective for reconstructing the major evolutionary relationships among the selected felid species.